

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 11

REVIEW Lecture 10:

- **Classification of Partial Differential Equations (PDEs) and examples with finite difference discretizations**
 - Parabolic PDEs
 - Elliptic PDEs
 - Hyperbolic PDEs
- **Error Types and Discretization Properties:** $\mathcal{L}(\phi) = 0$, $\hat{\mathcal{L}}_{\Delta x}(\hat{\phi}) = 0$
 - Consistency: $\left| \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\phi) \right| \rightarrow 0$ when $\Delta x \rightarrow 0$
 - Truncation error: $\tau_{\Delta x} = \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\phi) \rightarrow O(\Delta x^p)$ for $\Delta x \rightarrow 0$
 - Error equation: $\tau_{\Delta x} = \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\hat{\phi} + \varepsilon) = -\hat{\mathcal{L}}_{\Delta x}(\varepsilon)$ (for linear systems)
 - Stability: $\left\| \hat{\mathcal{L}}_{\Delta x}^{-1} \right\| < \text{Const.}$ (for linear systems)
 - Convergence: $\left\| \varepsilon \right\| \leq \left\| \hat{\mathcal{L}}_{\Delta x}^{-1} \right\| \left\| \tau_{\Delta x} \right\| \leq \alpha O(\Delta x^p)$

2.29 Numerical Fluid Mechanics

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REVIEW Lecture 10, Cont'd:

- Classification of PDEs and examples
- Error Types and Discretization Properties
- **Finite Differences based on Taylor Series Expansions**
 - Higher Order Accuracy Differences, with Examples
 - Incorporate more higher-order terms of the Taylor series expansion than strictly needed and express them as finite differences themselves (making them function of neighboring function values)
 - If these finite-differences are of sufficient accuracy, this pushes the remainder to higher order terms => increased order of accuracy of the FD method
 - General approximation:
$$\left(\frac{\partial^m u}{\partial x^m} \right)_j - \sum_{i=-r}^s a_i u_{j+i} = \tau_{\Delta x}$$
 - Taylor Tables or Method of Undetermined Coefficients (Polynomial Fitting)
 - Simply a more systematic way to solve for coefficients a_i

FINITE DIFFERENCES – Outline for Today

- Classification of Partial Differential Equations (PDEs) and examples with finite difference discretizations (Elliptic, Parabolic and Hyperbolic PDEs)
- **Error Types and Discretization Properties**
 - Consistency, Truncation error, Error equation, Stability, Convergence
- **Finite Differences based on Taylor Series Expansions**
 - Higher Order Accuracy Differences, with Example
 - Taylor Tables or Method of Undetermined Coefficients (Polynomial Fitting)
- **Polynomial approximations**
 - Newton's formulas
 - Lagrange polynomial and un-equally spaced differences
 - Hermite Polynomials and Compact/Pade's Difference schemes
 - Boundary conditions
 - Un-Equally spaced differences
 - Error Estimation: order of convergence, discretization error, Richardson's extrapolation, and iterative improvements using Roomberg's algorithm

References and Reading Assignments

- Chapter 23 on “Numerical Differentiation” and Chapter 18 on “Interpolation” of “Chapra and Canale, Numerical Methods for Engineers, 2006/2010/2014.”
- Chapter 3 on “Finite Difference Methods” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 3 on “Finite Difference Approximations” of “H. Lomax, T. H. Pulliam, D.W. Zingg, *Fundamentals of Computational Fluid Dynamics (Scientific Computation)*. Springer, 2003”

Finite Differences using Polynomial approximations

Numerical Interpolation:

“Historical” Newton’s Iteration Formula

Standard triangular family of polynomials

$$\begin{aligned} f(x) &= p(x) + r(x) \\ &= c_0 + c_1(x - x_0) + \cdots + c_n(x - x_0) \cdots (x - x_{n-1}) \\ &\quad + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0) \cdots (x - x_n) \end{aligned}$$

Divided Differences: $c_i = ?$

$$f(x_0) = c_0 \Rightarrow c_0 = \boxed{f(x_0)}$$

$$f(x_1) = c_0 + c_1(x_1 - x_0) \Rightarrow c_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \boxed{f[x_0, x_1]}$$

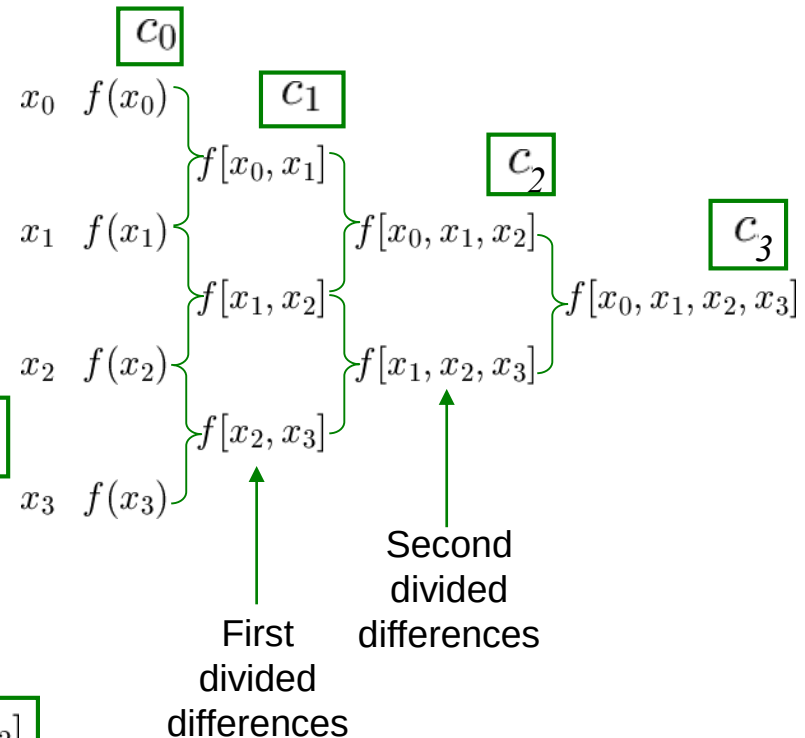
$$f(x_2) = c_0 + c_1(x_2 - x_0) + c_2(x_2 - x_0)(x_2 - x_1)$$

$$c_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0} = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \boxed{f[x_0, x_1, x_2]}$$

By recurrence:

$$\Rightarrow \boxed{c_n = f[x_0, x_1, \dots, x_n]} = \frac{f[x_1, \dots, x_n] - f[x_0, \dots, x_{n-1}]}{x_n - x_0}$$

Newton’s Computational Scheme



- Newton’s formula allow easy recursive computation of the coefficients of a polynomial of order n that interpolates $n+1$ data point
- Derivative of that polynomial can then be expressed as a function of these $n+1$ data points (in our case, unknown fct values)

Finite Differences using Polynomial approximations

Equidistant Newton's Interpolation

Equidistant Sampling

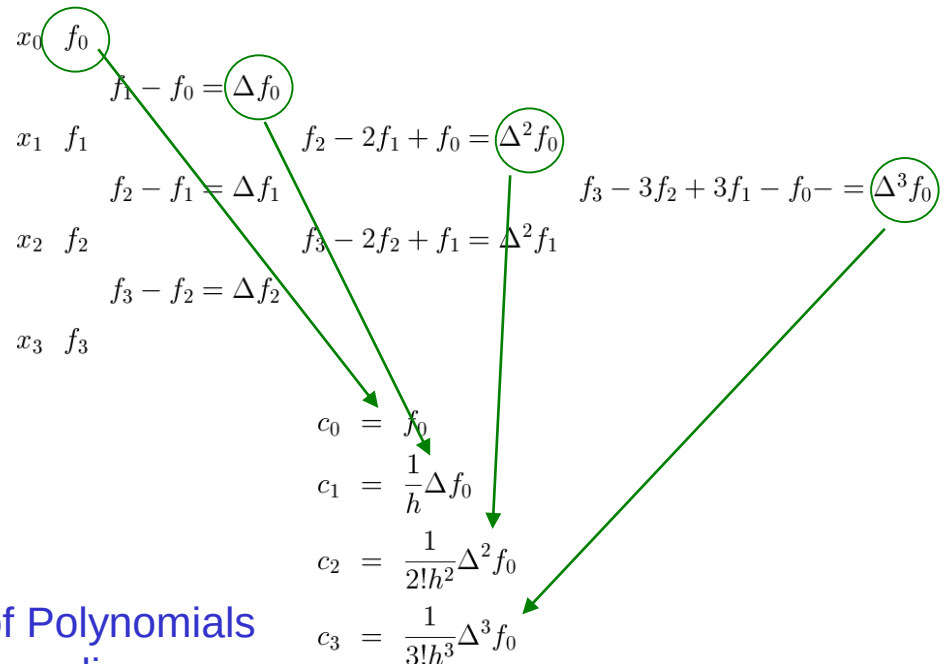
$$x_i = x_0 + ih$$

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1}{h}(f_1 - f_0) = \frac{1}{h} \Delta f_0$$

$$\begin{aligned} f[x_0, x_1, x_2] &= \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} \\ &= \frac{1}{1 \cdot 2 \cdot h^2}(f_2 - 2f_1 + f_0) = \frac{1}{2!h^2} \Delta^2 f_0 \end{aligned}$$

$$f[x_0, x_1, x_2, x_3] = \frac{1}{3! \cdot h^3}(f_3 - 3f_2 + 3f_1 - f_0) = \frac{1}{3!h^3} \Delta^3 f_0$$

Divided Differences with equidistant step size implied



Triangular Family of Polynomials Equidistant Sampling

$$\begin{aligned} f(x) &= f_0 + \frac{\Delta f_0}{h}(x - x_0) + \frac{\Delta^2 f_0}{2!h^2}(x - x_0)(x - x_1) + \cdots \\ &\quad + \frac{\Delta^n f_0}{n!h^n}(x - x_0)(x - x_1) \cdots (x - x_{n-1}) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0) \cdots (x - x_n) \end{aligned}$$

Numerical Differentiation using Newton's algorithm for equidistant sampling: 1st Order

First Derivatives

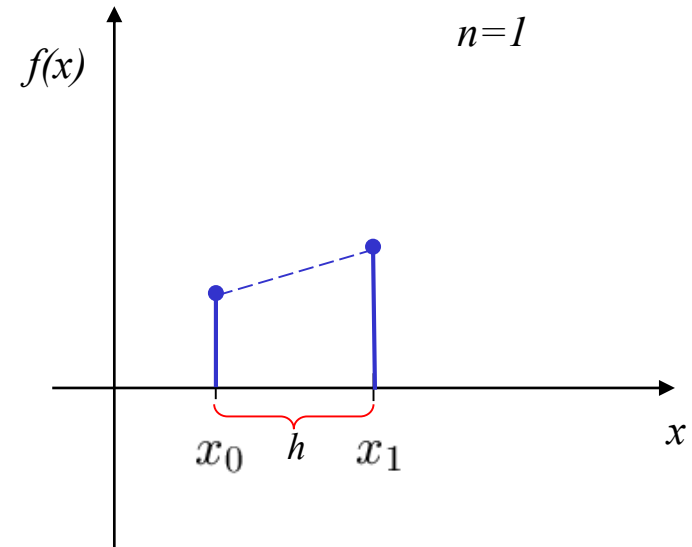
Triangular Family of Polynomials Equidistant Sampling

$$f(x) = f_0 + \frac{\Delta f_0}{h}(x - x_0) + \frac{\Delta^2 f_0}{2!h^2}(x - x_0)(x - x_1) + \cdots \\ + \frac{\Delta^n f_0}{n!h^n}(x - x_0)(x - x_1) \cdots (x - x_{n-1}) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0) \cdots (x - x_n)$$

First order
 $n = 1$

$$f(x) = f_0 + \frac{\Delta f_0}{h}(x - x_0) + \frac{f''(\xi)}{2!}(x - x_0)(x - x_1)$$

$$f'(x) = \frac{\Delta f_0}{h} + O(h) = \frac{1}{h}(f_1 - f_0) + O(h)$$



Numerical Differentiation using Newton's algorithm for equidistant sampling: 2nd Order

Second order

$$n = 2$$

$$f(x) = f_0 + \frac{\Delta f_0}{h}(x-x_0) + \frac{\Delta^2 f_0}{2!h^2}(x-x_0)(x-x_1) + \frac{f'''(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2) + \dots$$

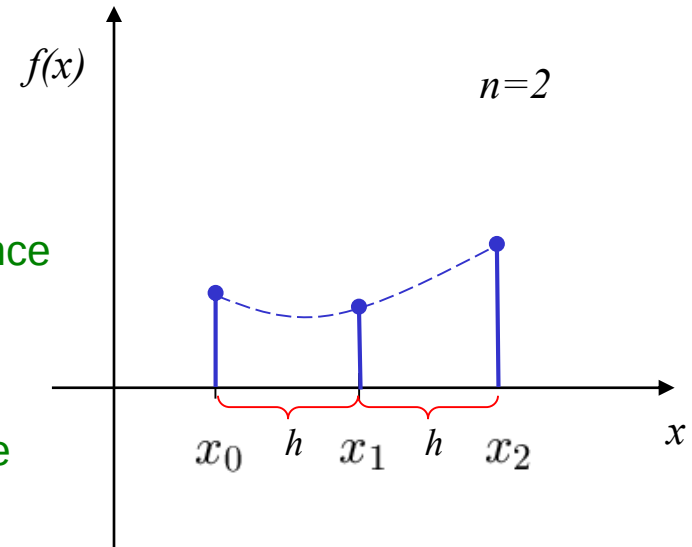
$$f'(x) = \frac{\Delta f_0}{h} + \frac{\Delta^2 f_0}{2h^2}(x-x_0) + \frac{\Delta^2 f_0}{2h^2}(x-x_1) + O(h^2)$$

$$\begin{aligned} f'(x_0) &= \frac{f_1 - f_0}{h} - \frac{1}{2h}(f_2 - 2f_1 + f_0) + O(h^2) \\ &= \frac{2f_1 - 2f_0 - f_2 + 2f_1 - f_0}{2h} + O(h^2) \\ &= \boxed{\frac{1}{h}\left(-\frac{3}{2}f_0 + 2f_1 - \frac{1}{2}f_2\right) + O(h^2)} \end{aligned}$$

Forward Difference

$$\begin{aligned} f'(x_1) &= \frac{f_1 - f_0}{h} + \frac{1}{2h}(f_2 - 2f_1 + f_0) + O(h^2) \\ &= \boxed{\frac{1}{2h}(f_2 - f_0) + O(h^2)} \end{aligned}$$

Central Difference



Second Derivatives

$$n=2 \quad f''(x_0) = \frac{\Delta^2 f_0}{h^2} + O(h) = \boxed{\frac{1}{h^2}(f_0 - 2f_1 + f_2) + O(h)} \quad \text{Forward Difference}$$

$$n=3 \quad f''(x_1) = \boxed{\frac{1}{h^2}(f_0 - 2f_1 + f_2) + O(h^2)} \quad \text{Central Difference}$$

Finite Differences using Polynomial approximations

Numerical Interpolation: Lagrange Polynomials

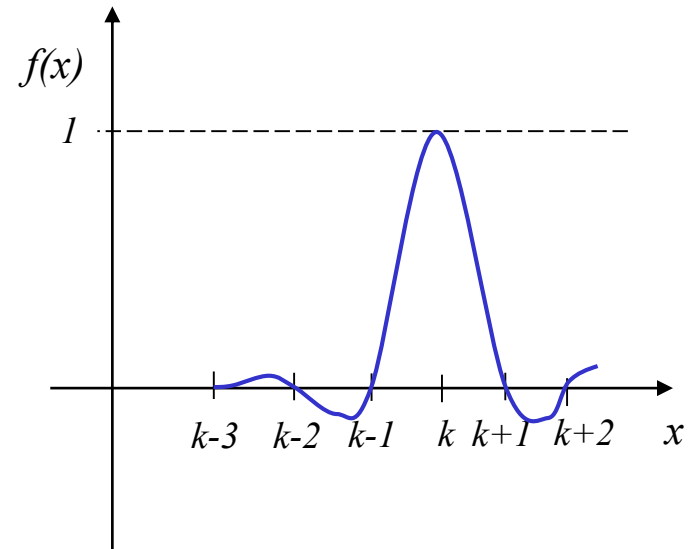
(Reformulation of Newton's polynomial)

$$p(x) = \sum_{k=0}^n L_k(x) f(x_k) = \sum_{k=0}^n L_k(x) f_k$$

$$L_k(x) = \sum_{i=0}^n \ell_{ik} x^i$$

$$L_k(x_i) = \delta_{ki} = \begin{cases} 0 & k \neq i \\ 1 & k = i \end{cases}$$

$$L_k(x) = \prod_{j=0, j \neq k}^n \frac{x - x_j}{x_k - x_j}$$



Difficult to program
Difficult to estimate errors
Divisions are expensive

Important for numerical integration
Nodal basis in FE

Hermite Interpolation Polynomials and Compact / Pade' Difference Schemes

- Use the values of the function and its derivative(s) at given points k
 - For example, for values of the function and of its first derivatives at pts k

$$u(x) = \sum_{k=1}^n a_k(x) u_k + \sum_{k=1}^m b_k(x) \left(\frac{\partial u}{\partial x} \right)_k$$

- General form for implicit/explicit schemes (here focusing on space)

$$\sum_{i=-r}^s b_i \left(\frac{\partial^m u}{\partial x^m} \right)_{j+i} - \sum_{i=-p}^q a_i u_{j+i} = \tau_{\Delta x}$$

- Generalizes the Lagrangian approach by using Hermitian interpolation
- Leads to the “Compact difference schemes” or “Pade' schemes”
- Are implemented by the use of efficient banded solvers
- Derivatives are then also unknowns

FINITE DIFFERENCES: Higher Order Accuracy

Taylor Tables for Pade' schemes

Table 3.3. Taylor table for central 3-point Hermitian approximation to a first derivative

$$d \left(\frac{\partial u}{\partial x} \right)_{j-1} + \left(\frac{\partial u}{\partial x} \right)_j + e \left(\frac{\partial u}{\partial x} \right)_{j+1} - \frac{1}{\Delta x} (a u_{j-1} + b u_j + c u_{j+1}) = ?$$

—	u_j	$\Delta x \left(\frac{\partial u}{\partial x} \right)_j$	$\Delta x^2 \left(\frac{\partial^2 u}{\partial x^2} \right)_j$	$\Delta x^3 \left(\frac{\partial^3 u}{\partial x^3} \right)_j$	$\Delta x^4 \left(\frac{\partial^4 u}{\partial x^4} \right)_j$	$\Delta x^5 \left(\frac{\partial^5 u}{\partial x^5} \right)_j$
$\Delta x d \left(\frac{\partial u}{\partial x} \right)_{j-1}$	—	d	$d \cdot (-1) \cdot \frac{1}{1}$	$d \cdot (-1)^2 \cdot \frac{1}{2}$	$d \cdot (-1)^3 \cdot \frac{1}{6}$	$d \cdot (-1)^4 \cdot \frac{1}{24}$
$\Delta x \left(\frac{\partial u}{\partial x} \right)_j$	—	1	—	—	—	—
$\Delta x e \left(\frac{\partial u}{\partial x} \right)_{j+1}$	—	e	$e \cdot (1) \cdot \frac{1}{1}$	$e \cdot (1)^2 \cdot \frac{1}{2}$	$e \cdot (1)^3 \cdot \frac{1}{6}$	$e \cdot (1)^4 \cdot \frac{1}{24}$
$-a \cdot u_{j-1}$	-a	$-a \cdot (-1) \cdot \frac{1}{1}$	$-a \cdot (-1)^2 \cdot \frac{1}{2}$	$-a \cdot (-1)^3 \cdot \frac{1}{6}$	$-a \cdot (-1)^4 \cdot \frac{1}{24}$	$-a \cdot (-1)^5 \cdot \frac{1}{120}$
$-b \cdot u_j$	-b	—	—	—	—	—
$-c \cdot u_{j+1}$	-c	$-c \cdot (1) \cdot \frac{1}{1}$	$-c \cdot (1)^2 \cdot \frac{1}{2}$	$-c \cdot (1)^3 \cdot \frac{1}{6}$	$-c \cdot (1)^4 \cdot \frac{1}{24}$	$-c \cdot (1)^5 \cdot \frac{1}{120}$

Image by MIT OpenCourseWare.

FINITE DIFFERENCES: Higher Order Accuracy

Taylor Tables for Pade' schemes, Cont'd

Table 3.3. Taylor table for central 3-point Hermitian approximation to a first derivative

$$\alpha \left(\frac{\partial \phi}{\partial x} \right)_{i+1} + \left(\frac{\partial \phi}{\partial x} \right)_i + \alpha \left(\frac{\partial \phi}{\partial x} \right)_{i-1} = \beta \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x} + \gamma \frac{\phi_{i+2} - \phi_{i-2}}{4\Delta x}$$

Image by MIT OpenCourseWare.

Sum each column starting from left and force the sums to be zero by proper choice of a, b, c, etc:

$$\begin{bmatrix} -1 & -1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 1 \\ -1 & 0 & -1 & -2 & 2 \\ 1 & 0 & -1 & 3 & 3 \\ -1 & 0 & -1 & -4 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow [a \quad b \quad c \quad d \quad e] = \frac{1}{4} [-3 \quad 0 \quad 3 \quad 1 \quad 1]$$

Truncation error is sum of the first column that does not vanish in the table, here 6th column (divided by Δx):

$$\tau_{\Delta x} = \frac{\Delta x^4}{120} \left(\frac{\partial^5 u}{\partial x^5} \right)_j$$

Compact / Pade' Difference Schemes: Examples

We can derive family of compact centered approximations for ϕ up to 6th order using:

$$\alpha \left(\frac{\partial \phi}{\partial x} \right)_{i+1} + \left(\frac{\partial \phi}{\partial x} \right)_i + \alpha \left(\frac{\partial \phi}{\partial x} \right)_{i-1} = \beta \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x} + \gamma \frac{\phi_{i+2} - \phi_{i-2}}{4\Delta x}$$

Scheme	Truncation error	α	β	γ
CDS-2	$\frac{(\Delta x)^2}{3!} \frac{\partial^3 \phi}{\partial x^3}$	0	1	0
CDS-4	$\frac{13(\Delta x)^4}{3 \cdot 3!} \frac{\partial^5 \phi}{\partial x^5}$	0	$\frac{4}{3}$	$-\frac{1}{3}$
Padé-4	$\frac{(\Delta x)^4}{5!} \frac{\partial^5 \phi}{\partial x^5}$	$\frac{1}{4}$	$\frac{3}{2}$	0
Padé-6	$\frac{4(\Delta x)^6}{7!} \frac{\partial^7 \phi}{\partial x^7}$	$\frac{1}{3}$	$\frac{14}{9}$	$\frac{1}{9}$

Comments:

- Pade' schemes use fewer computational nodes and thus are more compact than CDS
- Can be advantageous (more banded systems!)

Image by MIT OpenCourseWare.

Higher-Order Finite Difference Schemes

Considerations

- Retaining more terms in Taylor Series or in polynomial approximations allows to obtain FD schemes of increased order of accuracy
- However, higher-order approximations involve more nodes, hence more complex system of equations to solve and more complex treatment of boundary condition schemes
- Results shown for one variable still valid for mixed derivatives
- To approximate other terms that are not differentiated: reaction terms, etc
 - Values at the center node is normally all that is needed
 - However, for strongly nonlinear terms, care is needed (see later)
- Boundary conditions must be discretized

Finite Difference Schemes: Implementation of Boundary conditions

- For unique solutions, information is needed at boundaries
- Generally, one is given either:

- i) the variable: $u(x = x_{\text{bnd}}, t) = u_{\text{bnd}}(t)$ (Dirichlet BCs)
- ii) a gradient in a specific direction, e.g.: $\left. \frac{\partial u}{\partial x} \right|_{(x_{\text{bnd}}, t)} = \phi_{\text{bnd}}(t)$ (Neumann BCs)
- iii) a linear combination of the two quantities (Robin BCs)

- Straightforward cases:
 - If value is known, nothing special needed (one doesn't solve for the BC)
 - If derivatives are specified, for first-order schemes, this is also straightforward to treat

Finite Difference Schemes:

Implementation of Boundary conditions, Cont'd

- Harder cases: when higher-order approximations are used
 - At and near the boundary: nodes outside of domain would be needed
- Remedy: use different approximations at and near the boundary
 - Either, approximations of lower order are used
 - Or, approximations go deeper in the interior and are one-sided. For example,

- 1st order forward-difference: $\left. \frac{\partial u}{\partial x} \right|_{(x_{\text{bnd}}, t)} = 0 \Rightarrow \frac{u_2 - u_1}{x_2 - x_1} \approx 0 \Rightarrow u_1 = u_2$

- Parabolic fit to the bnd point and two inner points:

$$\left. \frac{\partial u}{\partial x} \right|_{(x_{\text{bnd}}, t)} \approx \frac{-u_3(x_2 - x_1)^2 + u_2(x_3 - x_1)^2 - u_1[(x_3 - x_1)^2 - (x_2 - x_1)^2]}{(x_2 - x_1)(x_3 - x_1)(x_3 - x_2)} \quad \left(\approx \frac{-u_3 + 4u_2 - 3u_1}{2\Delta x} \text{ for equidistant nodes} \right)$$

- Cubic fit to 4 nodes (3rd order difference): $\left. \frac{\partial u}{\partial x} \right|_{(x_{\text{bnd}}, t)} \approx \frac{2u_4 - 9u_3 + 18u_2 - 11u_1}{6\Delta x} + O(\Delta x^3) \text{ for equidistant nodes}$

- Compact schemes, cubic fit to 4 pts: $u_{(x_{\text{bnd}}, t)} = u_1 \approx \frac{18u_2 - 9u_3 + 2u_4}{11} - \frac{6\Delta x}{11} \left(\frac{\partial u}{\partial x} \right)_1 \text{ for equidistant nodes}$

- In Open-boundary systems, boundary problem is not well posed =>
 - Separate treatment for inflow/outflow points, multi-scale (embedded) approach and/or generalized inverse problem (using data in the interior)

Finite-Differences on Non-Uniform Grids: 1-D

- Truncation error depends not only on grid spacing but also on the derivatives of the variable

$$f(x_{i+1}) = f(x_i) + \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) + \frac{\Delta x^3}{3!} f'''(x_i) + \dots + \frac{\Delta x^n}{n!} f^n(x_i) + R_n$$

$$R_n = \frac{\Delta x^{n+1}}{n+1!} f^{(n+1)}(\xi)$$

- Uniform error distribution can not be achieved on a uniform grid => non-uniform grids
 - Use smaller (larger) Δx in regions where derivatives of the function are large (small) => uniform discretization error
 - However, in some approximation (centered-differences), specific terms cancel only when the spacing is uniform
- Example: Lets define $\Delta x_{i+1} = x_{i+1} - x_i$, $\Delta x_i = x_i - x_{i-1}$ and write the Taylor series at x_i :

$$f(x) = f(x_i) + (x - x_i) f'(x_i) + \frac{(x - x_i)^2}{2!} f''(x_i) + \frac{(x - x_i)^3}{3!} f'''(x_i) + \dots + \frac{(x - x_i)^n}{n!} f^n(x_i) + R_n$$

$$R_n = \frac{(x - x_i)^{n+1}}{n+1!} f^{(n+1)}(\xi)$$

Non-Uniform Grids Example: 1-D Central-difference

- Evaluate $f(x)$ at x_{i+1} and x_{i-1} , subtract results, lead to central-difference

$$f(x_{i+1}) = f(x_i) + \Delta x_{i+1} f'(x_i) + \frac{\Delta x_{i+1}^2}{2!} f''(x_i) + \frac{\Delta x_{i+1}^3}{3!} f'''(x_i) + \dots + \frac{\Delta x_{i+1}^n}{n!} f^n(x_i) + R_n$$

$$- \quad f(x_{i-1}) = f(x_i) - \Delta x_i f'(x_i) + \frac{\Delta x_i^2}{2!} f''(x_i) - \frac{\Delta x_i^3}{3!} f'''(x_i) + \dots + \frac{(-\Delta x_i)^n}{n!} f^n(x_i) + R_n$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{x_{i+1} - x_{i-1}} - \underbrace{\frac{\Delta x_{i+1}^2 - \Delta x_i^2}{2! (x_{i+1} - x_{i-1})} f''(x_i) - \frac{\Delta x_{i+1}^3 + \Delta x_i^3}{3! (x_{i+1} - x_{i-1})} f'''(x_i) + \dots + R_n}_{\text{Truncation error } \tau_{\Delta x}}$$

$$(\Delta x_{i+1} + \Delta x_i = x_{i+1} - x_{i-1})$$

= Truncation error $\tau_{\Delta x}$

- For a non-uniform mesh, the leading truncation term is $O(\Delta x)$
 - The more non-uniform the mesh, the larger the 1st term in truncation error
 - If the grid contracts/expands with a constant factor r_e : $\Delta x_{i+1} = r_e \Delta x_i$
 - Leading truncation error term is: $\tau_{\Delta x}^{r_e} \approx \frac{(1-r_e) \Delta x_i}{2} f''(x_i)$
 - If r_e is close to one, the first-order truncation error remains small: this is good for handling any types of unknown function $f(x)$

Non-Uniform Grids Example: 1-D Central-difference

- What also matters is: “rate of error reduction as grid is refined”!
- Consider case where refinement is done by adding more grid points but keeping a constant ratio of spacing (geometric progression), i.e.

$$\Delta x_{i+1}^{2h} = r_{e,2h} \Delta x_i^{2h}$$

$$\Delta x_{i+1} = r_{e,h} \Delta x_i$$

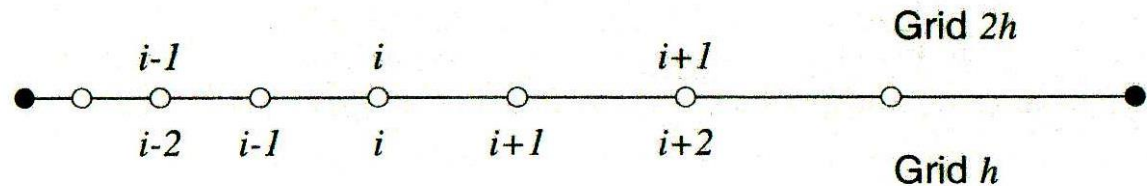


Fig. 3.3. Refinement of a non-uniform grid which expands by a constant factor r_e

- For coarse grid pts to be collocated with fine-grid pts: $(r_{e,h})^2 = r_{e,2h}$
- The ratio of the two truncation errors at a common point is then:

$$R \approx \frac{\frac{(1-r_{e,2h}) \Delta x_i^{2h}}{2} f''(x_i)}{\frac{(1-r_{e,h}) \Delta x_i^h}{2} f''(x_i)} \quad \text{which is} \quad \boxed{R \approx \frac{(1+r_{e,h})^2}{r_{e,h}}} \quad \text{since} \quad \underline{\Delta x_i^{2h} = \Delta x_i + \Delta x_{i-1} = (r_{e,h} + 1) \Delta x_{i-1}}$$

- The factor $R = 4$ if $r_e = 1$ (uniform grid). R is actually minimum at $r_e = 1$.
- When $r_e > 1$ (expanding grid) or $r_e < 1$ (contracting grid), the factor $R > 4$

Non-Uniform Grids Example: 1-D Central-difference

Conclusions

- When a non-uniform “geometric progression” grid is refined, error due to the 1st order term decreases faster than that of 2nd order term !
- Since $(r_{e,h})^2 = r_{e,2h}$, we have $r_{e,h} \rightarrow 1$ as the grid is refined. Hence, convergence becomes asymptotically 2nd order (1st order term cancels)
- Non-uniform grids are thus useful, if one can reduce Δx in regions where derivatives of the unknown solution are large
 - Automated means of adapting the grid to the solution (as it evolves)
 - However, automated grid adaptation schemes are more challenging in higher dimensions and for multivariate (e.g. physics-biology-acoustics) or multiscale problems
- (Adaptive) Grid generation still an area of active research in CFD
- Conclusions also valid for higher dimensions and for other methods (finite elements, etc)

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